

GIRISH KUMAR PATCHIKORU

Python Developer | Backend Software Engineer | Financial Services & Payment Systems

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PROFESSIONAL SUMMARY

- Python Developer and Software Engineer with **6 years of experience** working on backend applications, payment platforms, transaction-processing systems, and enterprise integrations across banking and financial services.
- Strong hands-on experience with **Python, FastAPI, Flask, REST APIs, SQL, and PySpark**, including backend services, transaction processing, reconciliation, data validation, automation, and integration workflows.
- Experienced with **real-time payments and card-processing applications**, including ACH, wire transfers, transaction status processing, payment validation, fraud-related workflows, and digital settlement systems.
- Worked with **Apache Kafka, IBM MQ, Redis, PostgreSQL, MySQL, SQL Server, and Db2** for asynchronous messaging, event processing, caching, and transaction-data handling.
- Comfortable working across **AWS and Microsoft Azure**, with practical experience using Docker, Kubernetes, Amazon EKS, OpenShift, Terraform, Azure DevOps, Git, and CI/CD pipelines.
- Regularly involved in **API development, testing, debugging, code reviews, release support, production troubleshooting, and incident analysis**, working with QA, DevOps, database, and application teams in Agile environments.

TECHNICAL SKILLS

- **Programming:** Python, Java, C#, JavaScript, SQL, PySpark
- **Python & Backend Development:** Python 3.x, FastAPI, Flask, Spring Boot, .NET Core, ASP.NET Web API, Node.js, REST APIs, OpenAPI, Microservices, JSON, Data Processing, Automation
- **Messaging & Real-Time Processing:** Apache Kafka, Kafka Producers/Consumers, IBM MQ, Event-Driven Architecture, Asynchronous Processing, Real-Time Transaction Processing, Change Data Capture (CDC)
- **Databases & Caching:** PostgreSQL, MySQL, SQL Server, Azure SQL, IBM Db2, Redis, Azure Cosmos DB, NoSQL
- **Cloud & Infrastructure:** AWS, Amazon EKS, AWS Lambda, Amazon VPC, Microsoft Azure, Azure Blob Storage, Azure Cognitive Search, Red Hat OpenShift
- **DevOps & Deployment:** Docker, Kubernetes, Terraform, Azure DevOps, CI/CD Pipelines, Git, Linux
- **API Security:** OAuth 2.0, Mutual TLS (mTLS), API Authentication, Access Control, Request Validation
- **Banking & Payments:** Credit Card Transactions, Real-Time Payments, ACH, Wire Transfers, Transaction Reconciliation, Payment Validation, Fraud Prevention Workflows, Core Banking Integration, Digital Settlement
- **AI & Data Processing:** Azure OpenAI Service, NLP, Document Processing, Text Extraction, Search & Summarization, PySpark.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Western Alliance Bank | Phoenix, Arizona.

Aug 2023 – present

- Worked on backend services for **WA Connect Live and Direct Connect APIs**, using Java, Spring Boot, and Node.js for real-time balance inquiries, payment requests, and commercial banking integrations.
- Built and updated **REST API endpoints** used for ACH transfers, wire payments, account information, and transaction-status requests from corporate and fintech applications.
- Added security controls around externally exposed APIs using **OAuth 2.0, mutual TLS, request validation, and access-control checks**.
- Worked with **PostgreSQL, MySQL, and Redis** for account, payment, and balance-related data, including caching frequently requested information used by backend services.
- Supported **Apache Kafka-based payment flows**, troubleshooting transaction events, status updates, failed messages, and downstream notifications between banking services.
- Helped move existing file-driven banking processes toward API-based workflows, handling validation rules, retries, error responses, and transaction-status mapping.
- Contributed to **Positive Pay and payment-verification services**, working with transaction exceptions, verification results, and status updates used in fraud-prevention workflows.
- Supported backend components for the **WA VenueX digital settlement platform**, working with Java/Spring Boot and Python services used in transfer, ledger, and treasury-processing workflows.
- Used **Python and PySpark** for transaction validation, reconciliation, data processing, and analytics tasks related to settlement and risk data.
- Worked with **PostgreSQL and NoSQL data stores** for settlement records, transaction information, audit events, and frequently accessed ledger data.
- Supported applications deployed across **AWS, Amazon EKS, Azure, Docker, and Kubernetes**, including configuration updates, health checks, log reviews, and application troubleshooting.
- Handled regular development activities such as **unit testing, API testing, code reviews, defect fixes, CI/CD deployments, release support, and production incident analysis** with QA, DevOps, and banking integration teams.

Tech Stack: Python, PySpark, Java, Spring Boot, Node.js, REST APIs, Open API, OAuth 2.0, mTLS, Apache Kafka, PostgreSQL, MySQL, Redis, NoSQL, AWS, Amazon EKS, AWS Lambda, Amazon VPC, Microsoft Azure, Azure DevOps, Kubernetes, Docker, Terraform, CI/CD, Git, Microservices, Event-Driven Architecture.

Software Engineer

Icertis Solution Ltd | Pune, India

Feb 2022 - Aug2023

- Worked on backend features for the **Icertis Contract Intelligence platform** using C#, .NET Core, ASP.NET Web API, and Entity Framework, including contract workflows, approvals, and internal service integrations.
- Built and modified **REST APIs** used for contract creation, metadata retrieval, document processing, and communication between application services.
- Wrote SQL queries and handled application data through **Azure SQL, SQL Server, and Entity Framework** for contract, workflow, and configuration-related operations.
- Used **Python, FastAPI, and Flask** for document ingestion, text preprocessing, metadata extraction, and lightweight services supporting contract-processing workflows.
- Supported **AI Studio document-processing features** by preparing and validating contract data used for document extraction and search workflows.
- Contributed to early **ExploreAI and Azure OpenAI work during 2023**, including API-flow testing, contract information retrieval, summarization, and search-related use cases.
- Worked with **Azure Blob Storage, Azure Cosmos DB, and Azure Cognitive Search** for storing contract documents, maintaining extracted metadata, and retrieving searchable contract information.

- Helped maintain enterprise integrations with **Salesforce, Microsoft Dynamics 365, SAP, and Office 365**, including API changes, field mapping, validation, and troubleshooting synchronization issues.
- Supported releases through **unit testing, API testing, bug fixes, log analysis, deployment checks, and production troubleshooting**, working with development, QA, and DevOps teams.

Tech Stack: Python 3.9/3.10, FastAPI, Flask, C#, .NET Core 3.1 / .NET 6, ASP.NET Web API, Entity Framework Core, REST APIs, SQL, SQL Server, Azure SQL, Microsoft Azure, Azure Blob Storage, Azure Cosmos DB, Azure Cognitive Search, Azure OpenAI Service, NLP, JavaScript, Git, CI/CD, Agile/Scrum, Contract Lifecycle Management (CLM)

Associate Software Engineer

Tata Consultancy Services | Hyderabad, India.

July 2020 - Feb 2022

- Worked on existing **Java and Spring Boot services** supporting core banking and payment flows, handling REST API changes, request validation, and smaller backend enhancements assigned during sprint development.
- Updated business logic around **payment validation, transaction status, account processing, and exception scenarios**, and helped with changes as parts of the application were moved into microservices.
- Supported **Kafka producers and consumers** used for asynchronous transaction processing, including checking message flow and investigating failed, delayed, or incorrectly processed events.
- Worked with **IBM MQ** where newer services exchanged messages with existing banking applications, helping with payload validation, queue-related issues, and failure troubleshooting.
- Wrote and modified **SQL queries in PostgreSQL and IBM Db2** to verify account, payment, and transaction data during development, testing, and defect investigation.
- Used **Redis** for frequently accessed reference data and helped investigate issues involving cached values, backend responses, and differences between application and database records.
- Assisted with application deployments and configuration changes across **AWS and Red Hat OpenShift**, reviewing logs, service health, and deployment issues together with development and DevOps teams.
- Spent regular sprint cycles on **unit testing, API testing, bug fixes, code reviews, regression checks, release support, and documentation**, working closely with QA, database, and senior engineering teams.

Tech Stack: Java 8/11, Spring Boot 2.x, REST APIs, Apache Kafka 2.6–2.8, IBM MQ 9.2, PostgreSQL 12/13, IBM Db2 11.5, Redis 6.x, AWS, Red Hat OpenShift, Docker, Kubernetes, Git, CI/CD, SQL, Linux, JSON, Microservices, Event-Driven Architecture.

EDUCATION

Bachelor of Engineering in Computer Science

Jawaharlal Nehru Technological University Kakinada